

7.1 $\mathbb{R}^{10}$ $[f]_B^B = \begin{pmatrix} J_{0,5} & & \\ & J_{0,3} & \\ & & J_{0,2} \end{pmatrix}$

$B = (\vec{b}_1, \dots, \vec{b}_{10})$
 $\hookrightarrow$ báze $\mathbb{R}^{10}$

• Pre $\forall i, j \in \mathbb{N}$ určí dim
 a pomocou báze B nejchnejších
 bázu podpr.
 $(\ker f^i) \cap (\operatorname{Im} f^j)$

$[f]_B^B = g$

• $(\ker g^i)$; pre $i \in \{1, \dots, 5\}$ platí, že v každom
 vektore bude prvých i vekt. v $(\ker g^i)$

5	4	3	2	1	0
•	•	•	•	•	•
		•	•	•	•
		•	•	•	•
			•	•	•
			•	•	•

• $(\operatorname{Im} g^j)$; pre $j \in \mathbb{N}$ platí, že v každom vekt. bude "posledných" j vekt. ležat v $(\operatorname{Im} g^j)$
 $\hookrightarrow j$ vekt. od zadn. vekt.

$\ker g = \operatorname{span}\{e_1, e_6, e_9\}$

$\ker g^2 = \operatorname{span}\{e_2, e_7, e_{10}\} + \ker g$

$\ker g^3 = \operatorname{span}\{e_3, e_8\} + \ker g^2$

$\ker g^4 = \operatorname{span}\{e_4\} + \ker g^3$

$\ker g^5 = \operatorname{span}\{e_5\} + \ker g^4 \Rightarrow$ pre $i \geq 5$ je $\ker g^i = \mathbb{R}^{10}$

$\operatorname{Im} g = \{e_{1-4}, e_{6-7}, e_9\} \rightarrow e_{1-4} := e_1, e_2, \dots, e_4$

$\operatorname{Im} g^2 = \{e_{1-3}, e_6\}$

$\operatorname{Im} g^3 = \{e_{1-2}\}$

$\operatorname{Im} g^4 = \{e_1\}$

$\operatorname{Im} g^5 = \{0\} \rightarrow$ pre $j \geq 5$ je $\operatorname{Im} g^j = \{0\}$

* $n_i = \dim(\ker f^i) \cap (\operatorname{Im} f^j)$ báza

• $(\ker f^i) \cap (\operatorname{Im} f^j)$: $\rightarrow g = [f]_B^B$, takže $[e_i]_B = b_1, \dots$
 $\rightarrow 5 \dots$ znamená, pre 5, 6, 7, 8... a ďalej (pre $\forall$ čísla $\in \mathbb{N}$ od 5)

• $j = 5 \dots, i = 1-5 \dots$ * $n = \{0\}$ $\dim(n) = 0$

• $j = 4, i = 1-5 \dots$ $n = \{b_1\}$ $\dim(n) = 1$

• $j = 3, i = 2-5 \dots$ $n = \{b_{1-2}\}$ $\dim(n) = 2$

$i = 1$ $n = \{b_1\}$ $\dim(n) = 1$

• $j = 2, i = 3-5 \dots$ $n = \{b_{1-3}, b_6\}$ $\dim(n) = 4$

$i = 2$ $n = \{b_{1-2}, b_6\}$ $\dim(n) = 3$

$i = 1$ $n = \{b_1, b_6\}$ $\dim(n) = 2$

$$j=1, i=5 \dots$$

$$i=4$$

$$i=3$$

$$i=2$$

$$i=1$$

$$n = (b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$$

$$n = (b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$$

$$n = (b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$$

$$n = (b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$$

$$n = (b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$$

$$\dim(n) = 8$$

$$-11- = 7$$

$$= 6$$

$$-111- = 5$$

$$= 3$$